

Zhiheng Zhang

m15601654187@163.com | zhihezhang@student.unimelb.edu.au
github.com/zhiheng-zhang-Mera

Melbourne, Australia

Research Profile

Computer Science researcher and research engineer focused on **trustworthy AI systems, ML-systems security, long-horizon execution, runtime reliability, and empirical evaluation.**

Education

University of Melbourne

Master's degree in Computer Science

Melbourne, Australia

Expected 2026

University of British Columbia

Bachelor of Science in Computer Science

Canada

Completed Summer 2024

Selected Research and Engineering Projects

Codex Boss — *Multi-agent orchestration and evidence-based acceptance platform*

2026–present

github.com/zhiheng-zhang-Mera/Codex-Boss

- Building a platform for long-running AI workflows in which multiple agents can plan, execute, critique, and adjudicate work while preserving durable task state.
- Developed platform foundations spanning durable state/events, capability boundaries, plugin isolation, knowledge/data lifecycle, dogfooding, and acceptance/verification.
- Investigating how autonomous systems can distinguish meaningful task completion from superficially plausible but weak evidence.
- Treating unresolved failure cases as evidence for further evaluation rather than reporting an unfinished acceptance mechanism as solved.

DS-Hns — *Long-horizon autonomous software-work runtime*

2026–present

github.com/zhiheng-zhang-Mera/DS-Hns

- Building an execution layer for unattended software work with optional capabilities isolated as extensions.
- Implemented ordered/scheduled task queues, hardware-adaptive local concurrency, unified task lifecycle and terminal events, task history, and terminal notifications.
- Designed extension-failure isolation, restart/recovery-oriented execution, and an idempotent Windows installer with explicit optional-plugin flow.
- Using the system as a testbed for persistent execution, failure recovery, scheduling, and completion verification in long-running software-agent workflows.

Privacy Lens — *Trustworthy-software / privacy prototype*

- Explored privacy-risk interpretation and user-facing warning mechanisms with explicit attention to evidence boundaries, provenance, auditability, and bounded claims.
- Reinforced a research practice of separating implementation evidence from stronger security, privacy, or compliance conclusions.

Research Interests

Trustworthy AI systems; ML systems security; runtime isolation and capability boundaries; long-horizon failure propagation; empirical security and reliability evaluation.

Technical and Research Strengths

- End-to-end research engineering: problem formulation, system design, iterative implementation, testing, debugging, and empirical evaluation.
- Multi-module software design with explicit persistence, lifecycle, extensibility, isolation, recovery, and acceptance criteria.
- Git/GitHub-based experimental development, branch-based iteration, regression testing, and reproducible project documentation.
- Practical multi-AI workflow design while independently reviewing plans, evidence, failure modes, and completion claims.

Current Research Direction

I am interested in **trustworthy AI systems under long-horizon execution**: how failures propagate when model outputs are coupled to runtime state, tools, permissions and subsequent actions, and how runtime boundaries, audit evidence and empirical evaluation can improve security and reliability without treating the model as an isolated component.